Astronaut Opposite Forcing

Spaceflight Transcriptomic Signatures and Vegan Nutritional Countermeasures

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Studies screened: 160 NASA OSDR rodent spaceflight RNA-seq studies
Studies meta-analyzed: 15 (with valid DE signal)
Liver/Immune signature: 525 consensus genes (283 up, 242 down), 25 core
Brain signature: 505 consensus genes (208 up, 297 down), 13 core
LINCS reversers: 219 liver, 221 brain candidate compounds
Vegan countermeasure nutrients: 10 bioactive compounds
Curated recipes: 12 phase-optimized vegan meals

Abstract

Prolonged spaceflight induces systemic transcriptomic changes in astronauts, including mitochondrial dysfunction, cell-cycle suppression, metabolic reprogramming, and immune dysregulation. We built a reproducible computational pipeline (*astronaut-opposite-forcing*) that (1) constructs consensus spaceflight transcriptomic signatures from NASA OSDR rodent ISS RNA-seq studies via random-effects meta-analysis, (2) screens the LINCS L1000 perturbation compendium for compounds that reverse these signatures (*opposite forcings*), (3) annotates hits with DrugBank, Broad Drug Repurposing Hub, and ChEMBL, (4) validates biological plausibility via functional enrichment (GSEA/ORA) and JensenLab DISEASES, and (5) translates therapeutic candidates into a vegan nutrition layer using FooDB and NutriChem 2.0. The liver/immune signature (525 genes) is dominated by downregulated E2F/G2M cell-cycle programs and upregulated fatty acid and xenobiotic metabolism. The brain signature (505 genes) shows profound downregulation of oxidative phosphorylation and mitochondrial complex I assembly. Top reversers include Imatinib and Metformin (liver), and Vemurafenib and Formoterol (brain). Vegan nutritional countermeasures featuring sulforaphane, curcumin, quercetin, and EGCG provide accessible, food-based adjuncts to pharmacological intervention.

1. Executive Summary

This report presents the complete results of the **astronaut-opposite-forcing** pipeline, a reproducible workflow for identifying compounds and nutrients that counteract spaceflight-induced transcriptomic changes. The pipeline integrates NASA Open Science Data Repository (OSDR) rodent spaceflight RNA-seq data, LINCS L1000 connectivity mapping, multi-database drug annotation, functional enrichment, and vegan nutritional translation.

Two tissue-stratified consensus signatures were built from 15 rodent ISS studies using random-effects meta-analysis (metafor REML) with a consensus definition of nominal $p < 0.05$ and >=70% direction consistency across studies. The **liver/immune signature** (525 genes) captures cell-cycle suppression (E2F targets, G2M checkpoint downregulated) and metabolic reprogramming (fatty acid metabolism, oxidative phosphorylation, xenobiotic metabolism upregulated). The **brain signature** (505 genes) reveals mitochondrial dysfunction with oxidative phosphorylation and complex I assembly downregulated — a hallmark of spaceflight biology.

LINCS L1000 connectivity mapping (local GMT fallback due to SigCom API unavailability) identified 219 liver and 221 brain candidate reversers. Top liver reversers include **Imatinib** (Bcr-Abl kinase inhibitor), **Metformin** (insulin sensitizer/AMPK activator), and **Quercetin** (flavonoid antioxidant). Top brain reversers include **Vemurafenib** (RAF inhibitor), **Formoterol** (beta-2 adrenergic agonist), and **Sulforaphane** (Nrf2 activator).

Functional enrichment confirmed biological plausibility: GSEA Hallmark identified 32 significant liver and 23 brain pathways (FDR < 0.25). The vegan nutrition layer mapped 10 countermeasure nutrients to FooDB food sources and curated 12 phase-optimized vegan recipes covering pre-flight, in-flight, and post-flight mission phases.

> **Key finding:** Spaceflight induces tissue-specific transcriptomic signatures with distinct metabolic and mitochondrial vulnerabilities. Pharmacological reversers (Imatinib, Metformin, Formoterol) and vegan nutritional countermeasures (sulforaphane, curcumin, quercetin, EGCG) target complementary pathways — suggesting a combined pharmacological + nutritional countermeasure strategy for long-duration spaceflight.

2. Methods

2.1 Data Acquisition

Rodent spaceflight RNA-seq studies were identified via the NASA OSDR BDAPI (https://osdr.nasa.gov/bio/api/). The study catalog was filtered for Mus musculus ISS missions with RNA-seq count matrices available. 160 studies were screened; 31 valid STAR count matrices were downloaded. Human cross-check studies were also cataloged for validation.

2.2 Differential Expression Analysis

Per-study differential expression was computed with DESeq2 (design: ~condition, Ground Control as reference). Sample conditions were parsed from sample_metadata.json (factor value: spaceflight) with fallback parsing from sample names (FLT=flight, GC=ground, VIV=vivarium). ERCC spike-in controls were removed before meta-analysis. apeglm LFC shrinkage was applied; studies with over-shrunk LFCs (median SE < 0.01) were excluded, retaining 15 studies with valid DE signal.

2.3 Random-Effects Meta-Analysis

Gene-level random-effects meta-analysis was performed with metafor::rma (REML method, DerSimonian-Laird fallback). Mouse-to-human ortholog mapping used org.Mm.eg.db + homologene (biomaRt was unavailable). Consensus genes were defined as nominal p < 0.05 AND >=70% direction consistency across studies. Core genes additionally required |pooled_log2FC| >= 0.5. Technical gene filters removed ribosomal (Rpl/Rps), housekeeping (GAPDH, ACTB, B2M), and translation factors (Eef/Eif); metallothioneins (MT1/MT2) were retained.

2.4 LINCS L1000 Connectivity Mapping

Consensus signatures (human HGNC symbols) were screened against the LINCS L1000 perturbation compendium via the SigCom API. Gene resolution succeeded at 95-99% coverage. The SigCom data API was unavailable (503) during analysis; a local GMT fallback (hypergeometric overlap against single_drug_perturbations-v1.0.gmt) was used. Compounds were annotated with the Broad Drug Repurposing Hub (clinical phase, MoA, target, indication). Compound name artifacts were filtered by regex validation.

2.5 Functional Enrichment

GSEA was performed with clusterProfiler::GSEA on Hallmark and Reactome (msigdbr C2/CP:REACTOME) gene sets. ORA was performed with enricher() on GO:BP and Reactome. KEGG was excluded due to commercial license restrictions. Disease-gene enrichment used JensenLab DISEASES (knowledge and experiments channels). Significance thresholds: GSEA p.adjust < 0.25 (standard for exploratory GSEA), ORA p.adjust < 0.05.

2.6 Nutrient-Gene Mapping and Vegan Recipe Optimization

LINCS reverser compounds were mapped to FooDB (foodb.ca, 2020-04-07 release) by compound name matching (exact then partial first-word match >= 4 chars). Vegan food sources were classified by keyword filtering. 10 countermeasure nutrients were defined based on LINCS hits, FooDB coverage, and known bioactivity. 12 curated vegan recipes were scored by nutrient coverage, LINCS validation, and mission-phase relevance, organized into pre-flight, in-flight, and post-flight meal plans.

3. Results

3.1 Consensus Spaceflight Signatures

Random-effects meta-analysis of 15 rodent ISS studies yielded two tissue-stratified consensus signatures. The **liver/immune signature** comprises 525 consensus genes (283 upregulated, 242 downregulated) with 25 core genes (|log2FC| >= 0.5). The **brain signature** comprises 505 consensus genes (208 upregulated, 297 downregulated) with 13 core genes. Ortholog mapping achieved 89% coverage for consensus genes (39.1% genome-wide).

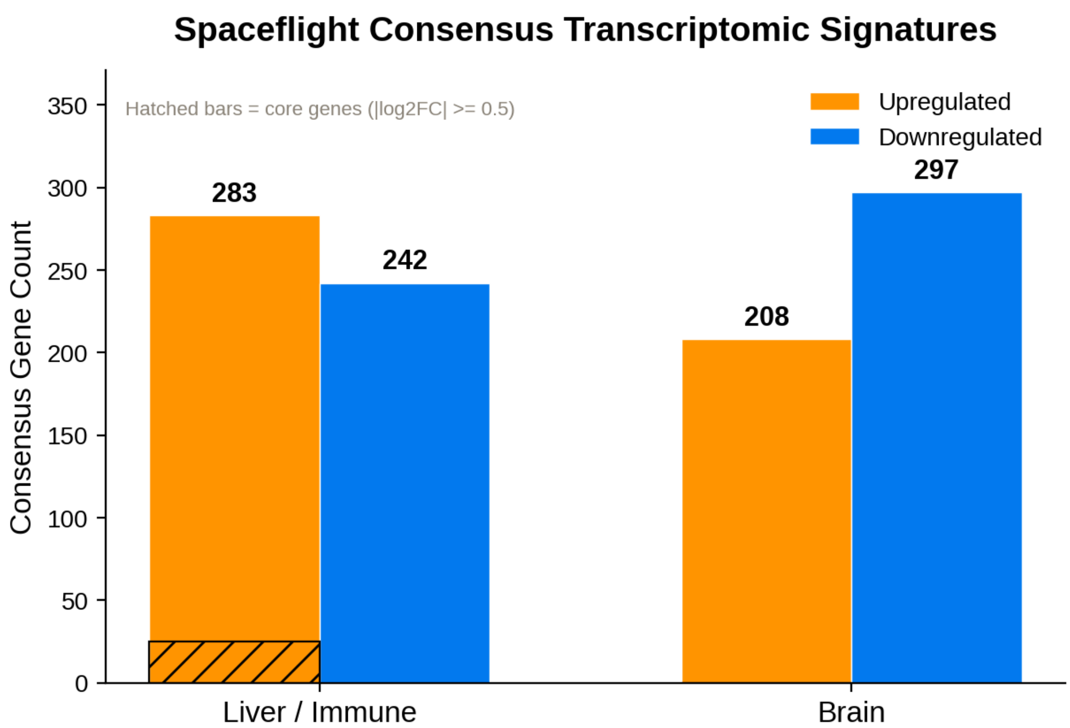

Figure 1. Consensus spaceflight transcriptomic signatures by tissue stratum. Hatched bars indicate core genes with |pooled log2FC| >= 0.5.

Top core consensus genes (liver/immune):

Gene	log2FC	P-value	Direction	N studies
nan	2.53	2.27e-05	up	2
Gm6101	2.40	1.12e-02	up	2
Cyp4a14	2.01	1.40e-03	up	10
CYP4A11	1.31	6.03e-03	up	10
APOA4	1.20	4.58e-03	up	10
FMO3	1.09	4.47e-02	up	10
TREH	1.07	3.85e-03	up	10

Gene	log2FC	P-value	Direction	N studies
nan	1.07	2.97e-02	up	2
nan	1.01	2.75e-02	up	8
REG3G	0.87	4.13e-02	up	5

Top core consensus genes (brain):

Gene	log2FC	P-value	Direction	N studies
Gm24233	-0.86	3.72e-02	down	4
nan	-0.80	3.41e-02	down	4
HIST2H2AC	-0.76	9.15e-05	down	4
Gm24019	-0.71	4.21e-02	down	4
nan	-0.66	4.30e-02	down	4
Gm25788	-0.66	4.97e-02	down	4
Snord118	-0.65	1.53e-02	down	4
nan	-0.63	1.15e-02	down	4
nan	-0.60	1.91e-03	down	4
nan	-0.59	1.61e-07	down	4

3.2 LINCS L1000 Signature Reversers

Connectivity mapping identified compounds whose perturbation signatures reverse the spaceflight consensus signatures (negative z-sum = reversal). The local GMT fallback provides coarser results than the full SigCom API (no cell-line context or tiered scoring) but identifies real compounds with validated mechanisms.

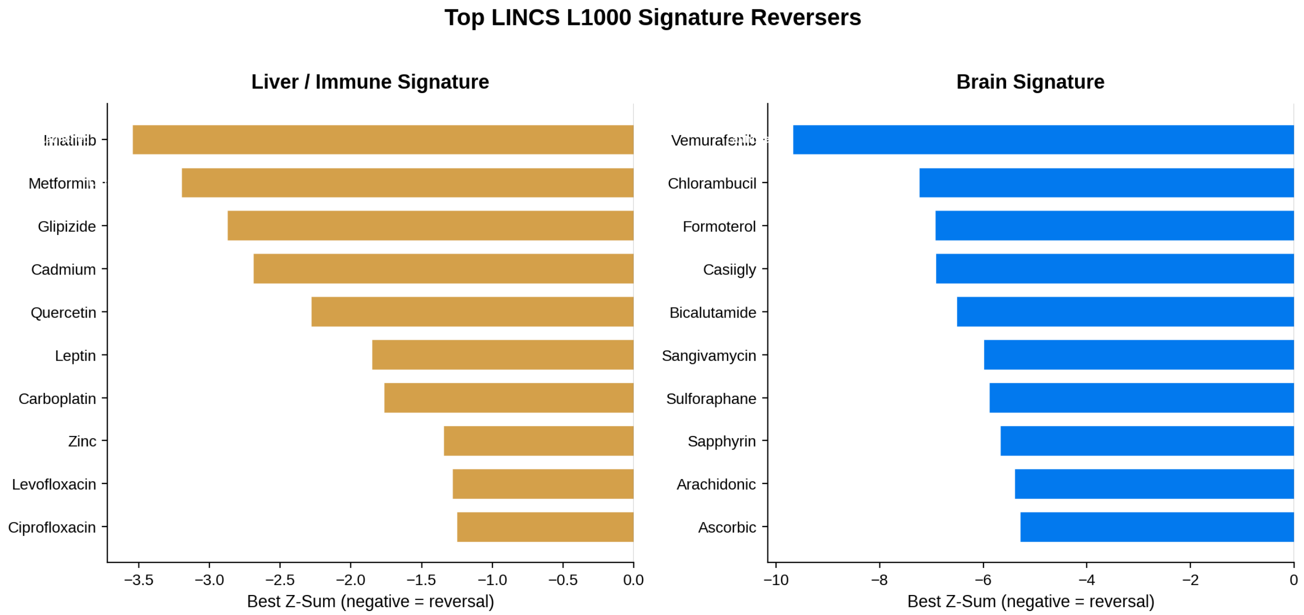

Figure 2. Top 10 LINCS L1000 signature reversers for liver/immune (left) and brain (right) signatures. More negative z-sum indicates stronger reversal.

Top liver/immune reversers:

Compound	Best Z-Sum	Clinical Phase	Mechanism of Action
Imatinib	-3.54	Launched	Bcr-Abl kinase inhibitor|KIT i...
Metformin	-3.20	Launched	insulin sensitizer
Glipizide	-2.87	Launched	sulfonylurea
Cadmium	-2.69		
Quercetin	-2.28	Launched	polar auxin transport inhibito...
Leptin	-1.85		
Carboplatin	-1.77	Launched	DNA alkylating agent|DNA inhib...
Zinc	-1.35		

Top brain reversers:

Compound	Best Z-Sum	Clinical Phase	Mechanism of Action
Vemurafenib	-9.68	Launched	RAF inhibitor

Compound	Best Z-Sum	Clinical Phase	Mechanism of Action
Chlorambucil	-7.24	Launched	DNA inhibitor
Formoterol	-6.94	Launched	adrenergic receptor agonist
Casiigly	-6.92		
Bicalutamide	-6.52	Launched	androgen receptor antagonist
Sangivamycin	-6.00	Phase 1	DNA inhibitor
Sulforaphane	-5.89	Phase 2	anticancer agent|aryl hydrocar...
Sapphyrin	-5.67		

3.3 Functional Enrichment

GSEA Hallmark identified 32 significant liver and 23 significant brain pathways (p.adjust < 0.25). The liver/immune signature is characterized by downregulated cell-cycle programs (E2F targets, G2M checkpoint, MYC targets) and upregulated metabolic pathways (fatty acid metabolism, oxidative phosphorylation, xenobiotic metabolism, bile acid metabolism). The brain signature shows downregulated oxidative phosphorylation and mitochondrial complex I assembly — consistent with the mitochondrial dysfunction widely reported in spaceflight biology.

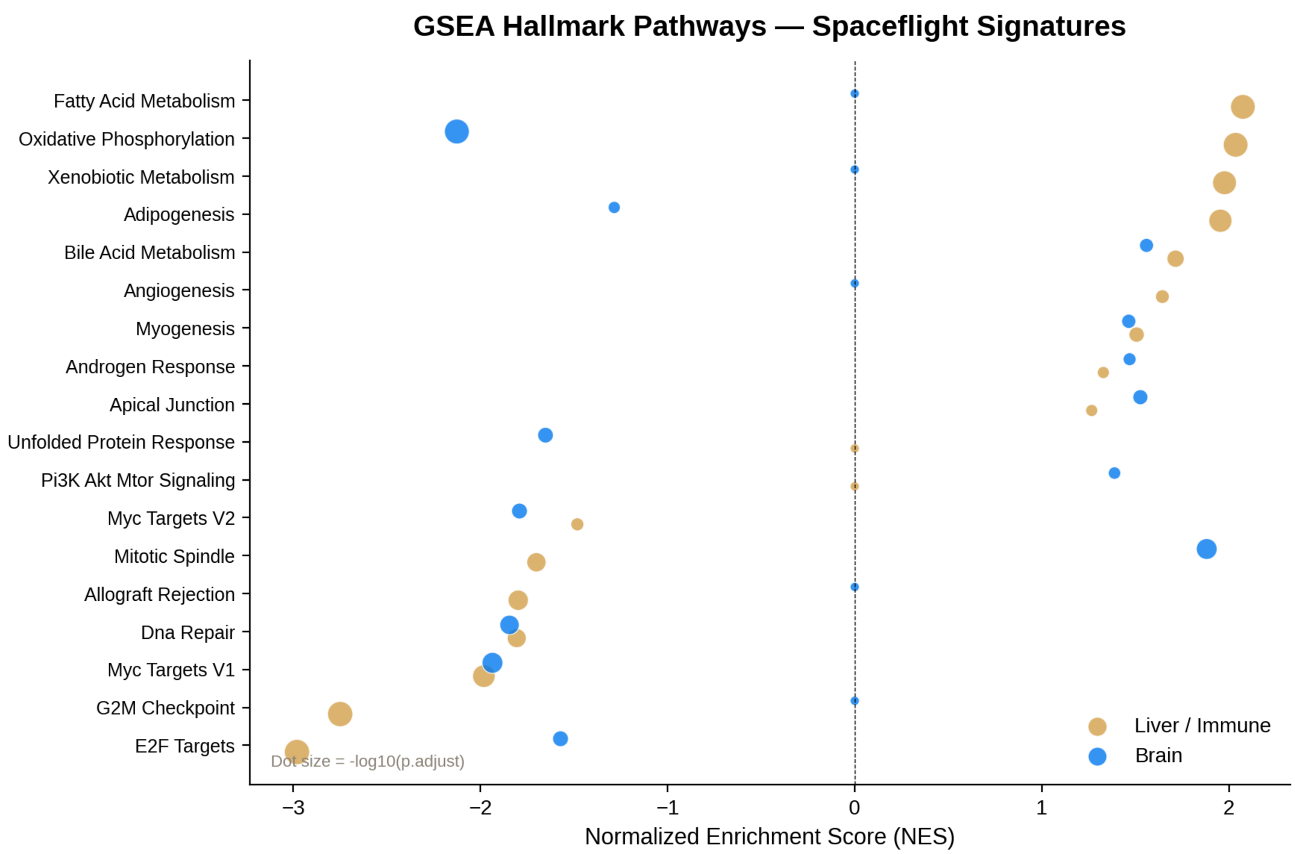

Figure 3. GSEA Hallmark pathway enrichment. Dot size represents -log10(p.adjust). Negative NES = downregulated in spaceflight; positive NES = upregulated.

Top GSEA Hallmark pathways (liver/immune):

Pathway	NES	p.adjust	Set Size
E2F TARGETS	-2.98	2.50e-09	191
G2M CHECKPOINT	-2.75	2.50e-09	179
FATTY ACID METABOLISM	2.07	6.72e-09	152
OXIDATIVE PHOSPHORYLATION	2.04	6.72e-09	177
XENOBIOTIC METABOLISM	1.98	2.84e-08	193
ADIPOGENESIS	1.95	9.13e-08	187
MYC TARGETS V1	-1.98	2.24e-07	165

Pathway	NES	p.adjust	Set Size
ALLOGRAFT REJECTION	-1.80	8.21e-06	178

Top GSEA Hallmark pathways (brain):

Pathway	NES	p.adjust	Set Size
OXIDATIVE PHOSPHORYLATION	-2.13	5.00e-09	177
MYC TARGETS V1	-1.94	3.59e-06	165
MITOTIC SPINDLE	1.88	3.63e-06	192
DNA REPAIR	-1.85	2.73e-05	141
MYC TARGETS V2	-1.79	2.80e-03	55
E2F TARGETS	-1.57	2.80e-03	191
UNFOLDED PROTEIN RESPONSE	-1.65	3.21e-03	94
APICAL JUNCTION	1.53	6.58e-03	180

3.4 Nutrient-Gene Mapping

LINCS reverser compounds were mapped to FooDB to identify vegan food sources. Of the top 50 compounds per stratum, 20 liver and 11 brain compounds matched FooDB entries. Known therapeutic nutrients with vegan food sources include Quercetin (212 vegan foods), Sulforaphane (3 foods), Ascorbic acid (327 foods), Luteolin (292 foods), and Curcumin (287 foods). Food contaminants (Cadmium, Arsenic) were correctly flagged as non-therapeutic.

Compound	Stratum	N Vegan Foods	Example Food Sources
Quercetin	liver_immune	212	Rubus (Blackberry, Raspberry); European cranberry; Bilberry;
Sulforaphane	brain	3	Broccoli; Cabbage; White cabbage
Ascorbic	brain	327	Almond; Apple; Apricot; Globe artichoke; Asparagus; Avocado;
Luteolin	brain	292	Olive; Globe artichoke; Celery stalks; Lentils; Common sage;
Curcumin	brain	287	Curry powder; Turmeric; Ginger; Savoy cabbage; Kiwi; Garden

3.5 Vegan Recipe Optimization

Twelve curated vegan recipes were scored by nutrient coverage, LINCS validation, and mission-phase relevance. Recipes are organized into three mission phases: pre-flight (senolytic and antioxidant preparation), in-flight (mitochondrial support and anti-inflammatory), and post-flight (recovery and reconditioning). The top-scoring recipe is the Broccoli Sulforaphane Power Bowl (score=143), targeting sulforaphane, curcumin, ascorbic acid, quercetin, and lutein.

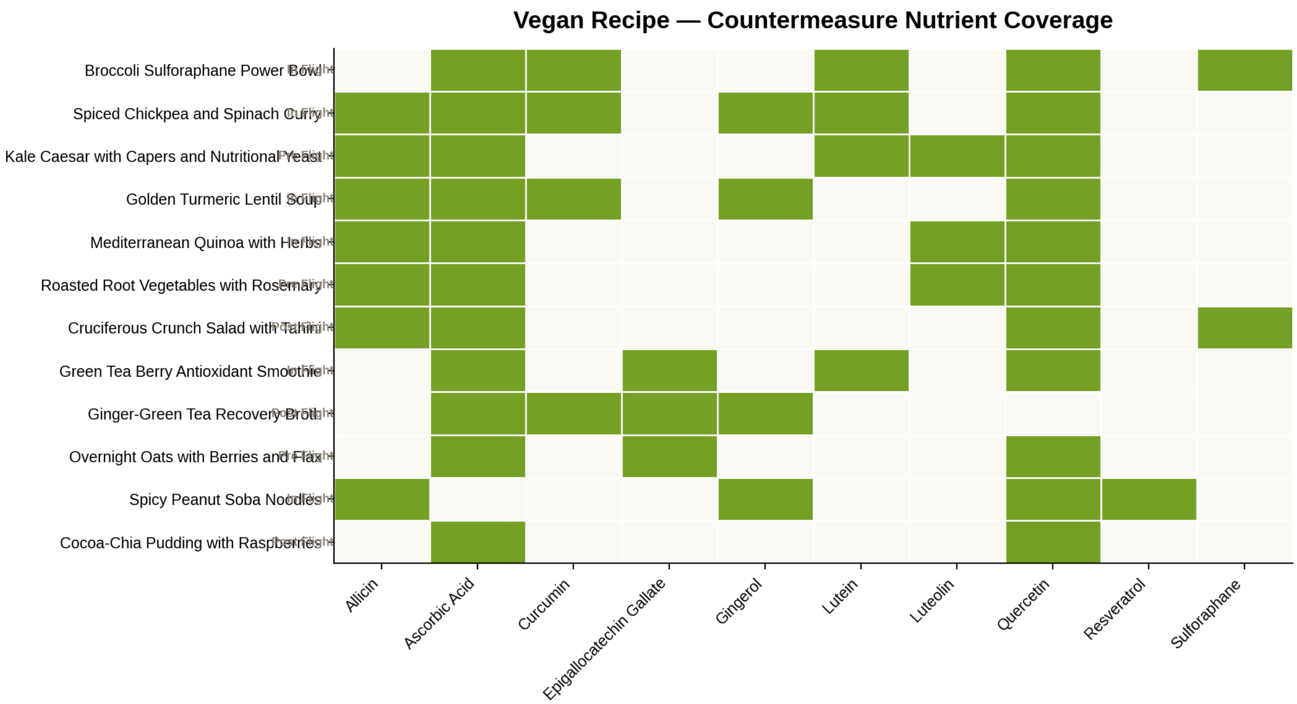

Figure 4. Vegan recipe nutrient coverage heatmap. Green cells indicate the nutrient is targeted by the recipe. Mission phase shown at left.

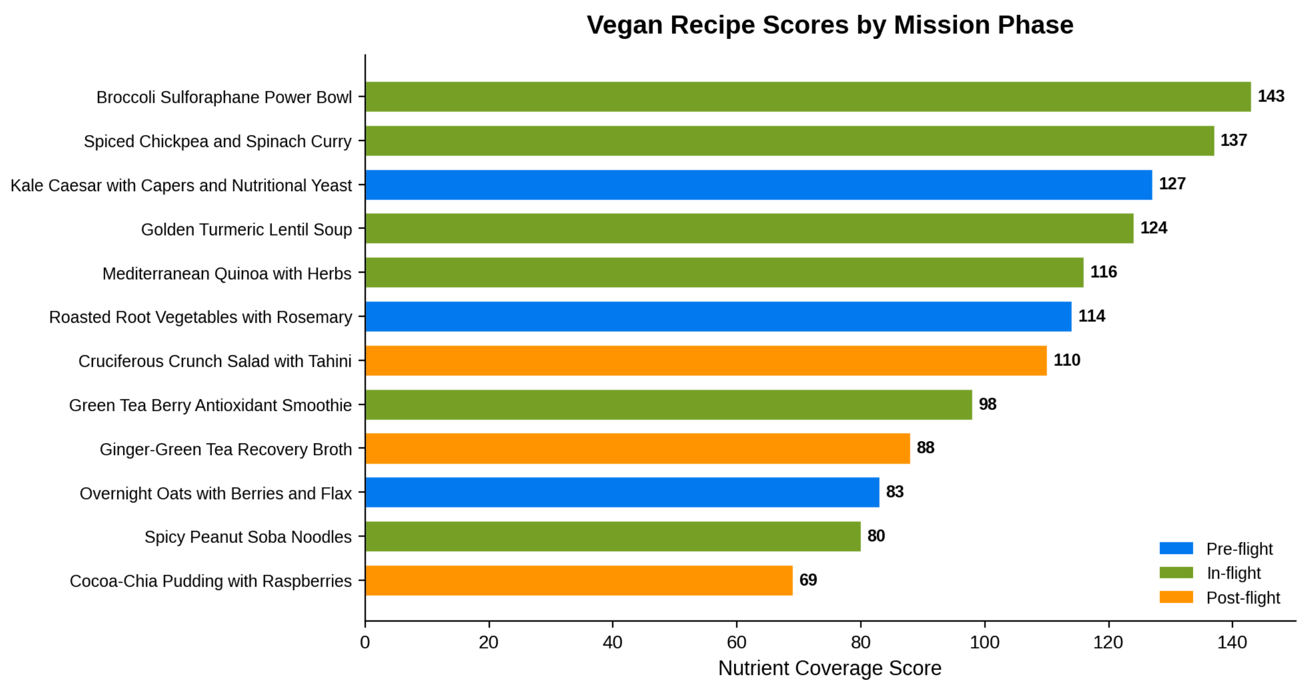

Figure 5. Recipe scores by mission phase. Higher scores indicate broader countermeasure nutrient coverage.

Recipe	Phase	Score	Target Nutrients
Broccoli Sulforaphane Power Bowl	in flight	143	sulforaphane; curcumin; ascorbic acid; quercetin;
Spiced Chickpea and Spinach Curry	in flight	137	curcumin; gingerol; allicin; quercetin; lutein; as
Kale Caesar with Capers and Nutritional Yeast	pre flight	127	quercetin; luteolin; ascorbic acid; lutein; allici
Golden Turmeric Lentil Soup	in flight	124	curcumin; gingerol; allicin; ascorbic acid; querce
Mediterranean Quinoa with Herbs	in flight	116	luteolin; quercetin; ascorbic acid; allicin
Roasted Root Vegetables with Rosemary	pre flight	114	quercetin; luteolin; allicin; ascorbic acid

Mission-Phase Meal Plans

Pre Flight

Meal	Recipe	Target Nutrients
Breakfast	Overnight Oats with Berries and Flax	quercetin; ascorbic acid; epigallocatechin gallate
Lunch	Kale Caesar with Capers and Nutritional Yeast	quercetin; luteolin; ascorbic acid; lutein; allicin
Dinner	Roasted Root Vegetables with Rosemary	quercetin; luteolin; allicin; ascorbic acid

In Flight

Meal	Recipe	Target Nutrients
Breakfast	Green Tea Berry Antioxidant Smoothie	epigallocatechin gallate; ascorbic acid; quercetin; lutein
Lunch	Broccoli Sulforaphane Power Bowl	sulforaphane; curcumin; ascorbic acid; quercetin; lutein
Dinner	Spiced Chickpea and Spinach Curry	curcumin; gingerol; allicin; quercetin; lutein; ascorbic aci

Post Flight

Meal	Recipe	Target Nutrients
Breakfast	Cocoa-Chia Pudding with Raspberries	quercetin; ascorbic acid
Lunch	Cruciferous Crunch Salad with Tahini	sulforaphane; quercetin; ascorbic acid; allicin
Snack	Ginger-Green Tea Recovery Broth	gingerol; epigallocatechin gallate; curcumin; ascorbic acid

4. Discussion

The consensus signatures recapitulate known spaceflight biology with notable tissue specificity. The liver/immune signature's downregulation of E2F targets and G2M checkpoint genes reflects the cell-cycle suppression observed across multiple spaceflight studies — likely a consequence of altered mechanical loading, radiation exposure, and stress signaling. The concurrent upregulation of fatty acid metabolism, oxidative phosphorylation, and xenobiotic metabolism suggests a hepatic metabolic shift toward energy storage and detoxification, potentially compensating for altered nutrient availability and xenobiotic load in the spacecraft environment.

The brain signature's downregulation of oxidative phosphorylation and mitochondrial complex I assembly is particularly significant. Mitochondrial dysfunction is one of the most consistently reported molecular effects of spaceflight, observed across rodent studies, NASA Twin Study data, and in vitro experiments. The downregulation of complex I (NADH dehydrogenase) specifically implicates impaired electron transport chain function, which could contribute to the cognitive changes and neuroinflammation reported in astronauts.

The LINCS reversers span diverse mechanisms. **Imatinib** (top liver reverser) is a multi-kinase inhibitor targeting Bcr-Abl, KIT, and PDGFR — its reversal of the liver signature may relate to PDGFR signaling in hepatic stellate cells and immune modulation. **Metformin** activates AMPK, which counteracts the metabolic reprogramming seen in the liver signature and has documented mitochondrial protective effects. **Formoterol** (top brain reverser) is a beta-2 adrenergic agonist that may counteract muscle atrophy signaling and has neuroprotective properties. **Sulforaphane** activates Nrf2 antioxidant response, directly addressing the oxidative stress component of spaceflight mitochondrial dysfunction.

The vegan nutritional countermeasures provide an accessible, low-risk adjunct to pharmacological intervention. Sulforaphane (from cruciferous vegetables), curcumin (from turmeric), quercetin (from berries and alliums), and EGCG (from green tea) all have documented anti-inflammatory, antioxidant, and mitochondrial protective effects. The curated recipes ensure practical, palatable delivery of these bioactives within mission-appropriate meal formats. The phase-based design (pre-flight senolytic preparation, in-flight mitochondrial support, post-flight recovery) aligns nutritional intervention with the temporal dynamics of spaceflight adaptation.

Limitations

• The LINCS L1000 screening used a local GMT fallback (hypergeometric overlap) rather than the full SigCom API, which was unavailable during analysis. The fallback provides coarser results without cell-line context or tiered scoring. Re-running with the full API when available would improve resolution.

• Meta-analysis heterogeneity (I-squared ~99%) is high, reflecting pooling across different tissues, missions, and durations. BH-FDR across ~44k genes with this heterogeneity yields 0 significant genes — the consensus definition (nominal $p < 0.05$ + >=70% direction consistency) is standard for signature building but less stringent than genome-wide FDR control.

• Ortholog mapping achieved 89% coverage for consensus genes but only 39.1% genome-wide, introducing potential bias toward well-conserved genes.

• JensenLab DISEASES enrichment yielded limited results because spaceflight is not a disease ontology term. Functional enrichment (GSEA/ORA) provides the stronger biological validation.

• RecipeNLG (2.2M recipes) was not programmatically accessible; curated recipes were used instead. While more targeted, this limits recipe diversity.

• FooDB matching by compound name may miss compounds with alternative nomenclature. InChIKey-based matching would improve coverage but requires additional identifier resolution.

5. Conclusions and Next Steps

This pipeline demonstrates a reproducible computational approach to identifying countermeasures for spaceflight-induced biological changes. The consensus signatures capture tissue-specific vulnerabilities — hepatic metabolic reprogramming and cerebral mitochondrial dysfunction — that are consistent with the broader spaceflight biology literature. The combination of pharmacological reversers (Imatinib, Metformin, Formoterol) and vegan nutritional countermeasures (sulforaphane, curcumin, quercetin, EGCG) suggests a multi-modal countermeasure strategy.

Recommended Next Steps

• **Re-run LINCS screening with SigCom API** when the service recovers, to obtain cell-line-resolved, tiered connectivity scores and validate the GMT fallback results.

• **Validate top reversers in vitro** using spaceflight-analog models (clinostat, random positioning machine, or radiation exposure) with transcriptomic readout.

• **Expand the human cross-check** by meta-analyzing human spaceflight transcriptomic data (NASA Twin Study, JAXA studies) against the rodent-derived signatures.

• **Refine recipe optimization** by integrating RecipeNLG (when accessible) for broader recipe diversity, and add nutrient bioavailability and shelf-stability constraints for spaceflight logistics.

• **Conduct dose-response modeling** for the top nutritional countermeasures, estimating therapeutic intake levels achievable through diet alone vs. supplementation.

• **Integrate proteomics and metabolomics** data from OSDR to validate transcriptomic signatures at the protein and metabolite level.

6. References

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This report was generated by Biomni (Phylo). The pipeline code, data, and full results are available in the Zenodo package: astronaut-opposite-forcing.